

LIM JACK SHENG

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EDUCATION

Bachelor of Computer Science (Computer System and Networking)	Oct 2022 - Present
Universiti Malaya	- CGPA 3.83
A - levels	Jul 2020 - Dec 2021
INTI International College Penang	- 3A, 1B
Secondary Education	Jan 2015 - Dec 2019
SMJK Keat Hwa	- SPM 11A

WORK EXPERIENCES

Software Developer Intern, ViTrox Technologies Sdn. Bhd.

Jul 2023 - Oct 2023

- Unit testing.
- Develop X-ray system features using Java and C++.
- Use localization.properties files to support multiple languages within the software application.
- Implement and utilize Gradle scripts to run unit tests written in C++ and generate results in HTML format.
- Design and develop the frontend interface for the machine using JavaFX.
- Technologies: **Java, C++, Gradle.**

Data Engineer Intern, RYT Bank

Jul 2024 - Jan 2025

- Maintained and optimized data pipelines in Apache Airflow to streamline data integration across multiple sources, including APIs, OSS storage, and PostgreSQL databases.
- Integrated Webhooks and Microsoft Power Automate on Microsoft Azure to enable notifications and alerts from Airflow directly to Microsoft Teams.
- Created a re-sync mechanism to ensure data consistency between BDT and Holistics.
- Developed an automated password re-check mechanism for data warehouse users, with alert email notifications for expired or invalid credentials.
- Implemented resource specification settings for task and manager pods within Kubernetes, optimizing resource allocation and management.
- Developed a housekeeping script for automated file and folder management and rolling deletion of files based on specific retention periods, using Bash and PowerShell.
- Deployed Spark events from Kafka to ArgoCD.
- Optimise Docker images on Alibaba Container Registry (ACR) using multi-stage build, reducing image size and memory usage by up to 80%.
- Monitored Airflow through Grafana and K9s to track the status of each pod and ensure smooth operations.
- Updated and configured Airflow write permissions to OSS buckets via Terraform.
- Contributed patches and updates to the Data Platform Helm repository, enhancing deployment stability and configuration consistency across environments.
- Improved Airflow performance by using .airflowignore to reduce DAG parsing time.
- Technologies: **Python, Airflow, Alibaba Cloud, Microsoft, DBT, Kubernetes, Docker, Bash, Powershell, Grafana, Prometheus.**

Data Engineer Intern, Grab

Jul 2025 - Dec 2025

- Standardizing timestamp formatting across different query engines to ensure consistent API behavior.

- Implemented Hystrix circuit breakers at strategic firing points to prevent database overload and improve system resilience during traffic spikes.
- Worked on prompt engineering tasks to generate high-quality metric documentation and evaluate the prompt via Langsmith.
- Contributed a new Forecast Tool to the MCP ecosystem for forecasting use cases.
- Designed and implemented GetMetricSimpleV2, a simplified API version that removes dependency on the query engine for faster and cleaner integration.
- Refactored the monolithic QueryMetric API into two clearer and more maintainable endpoints: GetMetricQuery and GetMetricValue.
- Enhanced the Internal SDK by adding support for one-sided filter expressions and improve the toString() output for filter DSL.
- Implementing derived metrics support in the Metric Availability API, including a new `latest timestamp` feature.
- Technologies: **Golang, AWS, Hystrix-Go, Datadog, Langsmith.**

PROJECTS

OpenSauce, Database Chatbot for Inventory Tracking

- Utilized LLM (Large Language Model) integration with a vector database for conducting similarity searches, enabling real-time tracking of inventory stocks with features for adding and updating basic stock information, and facilitating inventory inquiries through chatbot interaction.
- Technologies Used: **React, Firestore, Django, Pinecone, Gemini, PaLM.**

Medimate, AI-Powered Healthcare Assistant

- MediMate is an AI-powered healthcare assistant that enhances medication adherence, chronic disease management, and mental well-being. Key features include a voice-activated glucose checker, 24/7 Conversational AI, AI Health Transcriber, and Smart MedScan for instant medication registration via image capture. It also supports financial aid automation, integrates with wearables for real-time monitoring, and uses gamification to boost user engagement.
- Technologies Used: **React Native, Expo Go, Python, Firebase, Gemini AI, Google STT, ElevenLabs, Google Vision API, Google Fit API, Google Maps API.**

AmanahBlock, Blockchain-Powered for Charity Organizations

- AmanahBlock is a decentralized platform that combines blockchain technology, AI, and Shariah-compliant financial practices to transform the charitable giving ecosystem. Designed for transparency, inclusiveness, and impact, AmanahBlock addresses major inefficiencies and trust issues prevalent in traditional donation systems.
- Technologies Used: **React, Metamask, Privy integration, OpenAI, Gemini, BigChainDB, IPFS (Pinata), Solidity on Ethereum, OpenStreetMap API, HAQQ & Firoza Finance**

ACHIEVEMENTS

- 2nd runner-up, UM Hackathon 2023
- 2nd runner-up (Closed category), Programming League National 2023
- 2nd runner-up, KICT Innovative Hackathon (I-HAX) 2023
- 1st runner-up, UM Hackathon 2025
- 1st runner-up, KitaHack 2025
- Finalist, PayHack 2025

EXTRA-CURRICULAR ACTIVITIES

- UMHackathon 2024 - Director
- GDSC Developer Team Lead (2023 - 2024)
- GDSC Cloud Tech Team Core Team Member (2022 - 2023)